TRACK W (NONLINEAR WAVE): BLUEPRINT

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Abstract. A proof plan for track W: two-frequency quasi-periodic solutions of the cubic Dirichlet wave equation $u_{tt} = u_{xx} - mu - u^3$ on $[0, \pi]$ for every fixed $m > 0$, excited in the modes $J = \{1, 2\}$ (owner decision D30). Part 0 fixes every normalization once and proves milestone W0: the cubic attachment, the quartic coefficients, the partial Birkhoff normal-form action coefficients, nondegeneracy for every $m > 0$, and the Duffing consistency check. The proposed headline W-NLW2 closes the document. Nothing here is frozen or approved. The milestone plan is in `PLAN.md`.

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1. Conventions and normalizations

This part fixes every normalization of track W *once*. Later parts, the proposed headline (§3) and all future Lean statements use exactly these conventions; any departure must be declared as a deviation. Section 1.6 compares them with the printed conventions of the wave source, which are not internally consistent (§1.6, and `docs/tracks/W/errata-check-W0.md`).

1.1. Sources and locators.

- **P96-W**: J. Pöschel, *Quasi-periodic solutions for a nonlinear wave equation*, Comment. Math. Helv. **71** (1996), 269–296 (ETH E-Periodica scan). Printed page = PDF page +267.

- **P96-K**: J. Pöschel, *A KAM-theorem for some nonlinear partial differential equations*, Ann. Scuola Norm. Sup. Pisa (4) **23** (1996), 119–148 (Numdam scan). Printed page = PDF page +117.
- **KP96**: S. Kuksin, J. Pöschel, *Invariant Cantor manifolds of quasi-periodic oscillations for a nonlinear Schrödinger equation*, Ann. of Math. **143** (1996), 149–179; *author version 2.1*, cited by its own page numbers (not the journal's).

Versions and hashes are those of `docs/explorations/pde-kam-and-leapfrogging.md`, §1. Every formula of a scanned source relied on below was read on a page image, not only on the OCR layer.

1.2. **The equation.**

Standing assumption 1.1 (Mass)**.** $m > 0$ is fixed. All constants may depend on m unless stated otherwise; no uniformity as $m \to 0$ or $m \to \infty$ is claimed anywhere.

Definition 1.2 (Cubic Dirichlet wave equation)**.** A *classical solution* on an open interval $I \subset \mathbb{R}$ is a function $u : I \times [0,\pi] \to \mathbb{R}$ such that $u, u_t, u_x, u_{tt}, u_{xx}$ exist and are continuous on $I \times [0,\pi]$ (one-sided in x at the endpoints) and

$$u_{tt} = u_{xx} - mu - u^3 \quad \text{on } I \times [0,\pi], \qquad u(t,0) = u(t,\pi) = 0 \quad (t \in I). \tag{1}$$

Source: P96-W (1) and p. 272 with $f(u) = u^3$; the source writes $u_{tt} = u_{xx} - mu - f(u)$ with f odd, analytic, $f(u) = au^3 + O(u^5)$, $a \neq 0$; we take $a = 1$ and no higher terms (declared specialization).

Definition 1.3 (Dirichlet basis and linear frequencies)**.** For $j \geq 1$ let $\phi_j(x) = \sqrt{2/\pi}\,\sin(jx)$ and $\lambda_j = \sqrt{j^2 + m} > 0$. Then $\int_0^\pi \phi_i \phi_j \, dx = \delta_{ij}$, $-\phi_j'' + m\phi_j = \lambda_j^2 \phi_j$ and $\phi_j(0) = \phi_j(\pi) = 0$.

Source: P96-W p. 277, identical.

We record the elementary facts about λ used repeatedly: $j < \lambda_j < j + \sqrt{m}$, $j/\lambda_j < 1$, the map $j \mapsto \lambda_j$ is strictly increasing, the function $f(s) = \sqrt{s^2 + m}$ is strictly convex on $\mathbb{R}$ with $0 < f'(s) = s/f(s) < 1$ for $s > 0$, f' is strictly increasing, and $\lambda_{j+1} - \lambda_j = (2j+1)/(\lambda_{j+1} + \lambda_j) < 1$.

1.3. **Sequence spaces.**

Definition 1.4 (Weighted sequence spaces)**.** For $a \geq 0$ and $p \in \mathbb{R}$, $\ell^{a,p}$ is the space of real sequences $w = (w_j)_{j \geq 1}$ with

$$\|w\|_{a,p}^2 = \sum_{j \geq 1} |w_j|^2 \, j^{2p} e^{2aj} < \infty,$$

and $\ell^{a,p}_{\mathbb{C}}$ its complex analogue. The phase space is $E_{a,p} = \ell^{a,p} \times \ell^{a,p}$ with $\|(q,p)\|_E = \max(\|q\|_{a,p}, \|p\|_{a,p})$. The symbol p in $\ell^{a,p}$ is the regularity exponent; the second component of $E_{a,p}$ is the momentum, also written p; context separates them.

Source: P96-W p. 277 (the source's s is our p).

Convention 1.5 (Headline weights)**.** The headline and milestone W0 use $a = 1$ and $p = 1$: phase space $E_{1,1}$, nonlinear field in $\ell^{1,2}$. The source allows any $a > 0$, $s > 0$; this specialization removes two headline parameters. Lower exponents $\ell^{1,p}$, $p \leq 1$, appear only as target spaces of the unbounded linear part.

Definition 1.6 (Two-sided spaces)**.** For $a \geq 0$, $s \in \mathbb{R}$, $\ell^{a,s}_b$ is the space of complex sequences $(c_l)_{l \in \mathbb{Z}}$ with

$$\|c\|_{a,s}^2 = \sum_{l \in \mathbb{Z}} \langle l \rangle^{2s} e^{2a|l|} |c_l|^2 < \infty, \qquad \langle l \rangle = \max(1, |l|).$$

At $l = 0$ the weight is 1, so this is exactly the norm $|c_0|^2 + \sum_{l \neq 0} |c_l|^2 |l|^{2s} e^{2|l|a}$ of P96-W p. 279. The convolution is $(c * d)_l = \sum_{i \in \mathbb{Z}} c_i d_{l-i}$.

1.4. Coordinates, Hamiltonian and equations of motion.

Definition 1.7 (Reconstruction)**.** For $(q,p)\in E_{a,p}$ with $a>0$ put

$$U_m(q)(x)=\sum_{j\ge1}\frac{q_j}{\sqrt{\lambda_j}}\phi_j(x),\qquad V_m(p)(x)=\sum_{j\ge1}\sqrt{\lambda_j}\,p_j\phi_j(x).$$

The intended dictionary is $u=U_m(q)$, $u_t=V_m(p)$. Equivalently, if $u=\sum x_j\phi_j$ and $u_t=\sum y_j\phi_j$, then $q_j=\sqrt{\lambda_j}\,x_j$ and $p_j=y_j/\sqrt{\lambda_j}$; this is a symplectic rescaling, $dq_j\wedge dp_j=dx_j\wedge dy_j$.

Source: P96-W p. 277 ($u=\mathcal{S}q$, $v=\mathcal{S}'p$), identical.

Lemma 2.1 shows that for $a>0$ these series converge absolutely and uniformly on $[0,\pi]$ together with all their x-derivatives, for every $p\in\mathbb{R}$.

Definition 1.8 (Hamiltonian)**.** For $(q,p)\in E_{1,1}$,

$$H_m(q,p)=\Lambda(q,p)+G(q),\qquad \Lambda(q,p)=\frac12\sum_{j\ge1}\lambda_j\big(q_j^2+p_j^2\big),\qquad G(q)=\frac14\int_0^\pi U_m(q)(x)^4\,dx.$$

Λ is finite on $E_{a,p}$ for $p\ge\frac12$; in particular on $E_{1,1}$.

Source: P96-W (8) p. 280 and (9) p. 280 with $g(u)=u^4/4$; the energy $\frac12\langle v,v\rangle+\frac12\langle Au,u\rangle+\int g(u)$ of p. 277 in the coordinates of Definition 1.7.

Definition 1.9 (Cubic gradient)**.** For $q\in\ell^{1,1}$ let $G_q(q)$ be the sequence

$$G_q(q)_l=\frac{1}{\sqrt{\lambda_l}}\int_0^\pi U_m(q)(x)^3\phi_l(x)\,dx\qquad(l\ge1).$$

By Lemma 2.3, $G_q(q)_l=\partial G/\partial q_l$; by Theorem 2.5 it lies in $\ell^{1,2}$.

Definition 1.10 (Symplectic form and equations of motion)**.** The symplectic form is $\sum_j dq_j\wedge dp_j$ and the Hamiltonian equations are $\dot q_j=\partial H/\partial p_j$, $\dot p_j=-\partial H/\partial q_j$, i.e.

$$\dot q_j=\lambda_j p_j,\qquad \dot p_j=-\lambda_j q_j-G_q(q)_j\qquad(j\ge1).\tag{2}$$

The linear part $q\mapsto(\lambda_j q_j)$ maps $\ell^{a,p+1}$ boundedly to $\ell^{a,p}$ (norm $\le\sqrt{1+m}$) but not $\ell^{a,p}$ to itself; the system is read in $\ell^{1,0}$ for $(q,p)\in E_{1,1}$, or coordinatewise. The nonlinear part G_q is smoothing of order one (Theorem 2.5).

Source: P96-W (6) p. 277 and p. 280 ($X_G=\sum(\partial_{p_j}G\,\partial_{q_j}-\partial_{q_j}G\,\partial_{p_j})$), identical.

1.5. Complex coordinates, actions, angles.

Definition 1.11 (Complex coordinates)**.** $z_j=(q_j+ip_j)/\sqrt2$, $\bar z_j=(q_j-ip_j)/\sqrt2$, so $q_j=\sqrt2\,\mathrm{Re}\,z_j$, $p_j=\sqrt2\,\mathrm{Im}\,z_j$, $|z_j|^2=(q_j^2+p_j^2)/2$, and

$$\sum_j dq_j\wedge dp_j=i\sum_j dz_j\wedge d\bar z_j,\qquad \Lambda=\sum_j\lambda_j|z_j|^2.$$

Under the linear flow $\dot q_j=\lambda_j p_j$, $\dot p_j=-\lambda_j q_j$ one has $\dot z_j=-i\lambda_j z_j$.

Source: P96-W p. 281 (definition of z, form $i\sum dz\wedge d\bar z$, $\Lambda=\sum\lambda_j|z_j|^2$), identical; the recovery formula and the rotation sign printed later in P96-W differ, see §1.6.

Definition 1.12 (Actions and radian angles)**.** The (oscillator) action of mode j is $I_j=(q_j^2+p_j^2)/2=|z_j|^2$. On $\{I_j>0\}$ the radian angle φ_j is defined by

$$q_j=\sqrt{2I_j}\cos\varphi_j,\qquad p_j=-\sqrt{2I_j}\sin\varphi_j,\qquad z_j=\sqrt{I_j}\,e^{-i\varphi_j}.$$

Then $dq_j\wedge dp_j=d\varphi_j\wedge dI_j$ (Jacobian determinant 1), $\Lambda=\sum\lambda_jI_j$, and the linear flow is $\dot\varphi_j=\lambda_j$, $\dot I_j=0$. For tangential modes $j\in J=\{1,2\}$ we write $I=(I_1,I_2)$; for normal modes $j\ge3$ we write $Z_j=I_j$.

Convention 1.13 (Period-one torus angles and frequencies)**.** Torus parameters $\theta \in \mathbb{R}^2$ have period one: $\varphi_j = 2\pi\theta_j$. A torus with radian frequency vector ω (period 2π) has campaign frequency $\nu = \omega/(2\pi)$. The pair (θ, Y) with $Y = 2\pi I$ is canonical, $d\varphi \wedge dI = d\theta \wedge dY$. Small divisors in campaign units are $2\pi\, k \cdot \nu + l \cdot \Omega$. The linear torus of action $I \in (0,\infty)^2$ is

$$K_I^0(\theta) = (Q^0, P^0), \qquad Q_j^0 = \sqrt{2I_j}\cos(2\pi\theta_j), \quad P_j^0 = -\sqrt{2I_j}\sin(2\pi\theta_j) \quad (j = 1, 2),$$

and $Q_j^0 = P_j^0 = 0$ for $j \geq 3$, which is invariant under the linear flow with $\nu = (\lambda_1, \lambda_2)/(2\pi)$.

1.6. **Comparison with the printed conventions of the sources.** The following table maps each normalization to P96-W (and, where relevant, KP96). "Same" means checked identical on the page image. Details and derivations are in `docs/tracks/W/errata-check-W0.md`.

Object	This blueprint	P96-W (printed)
basis, λ_j	$\phi_j = \sqrt{2/\pi}\sin jx$, $\lambda_j = \sqrt{j^2+m}$	same (p. 277)
$\ell^{a,p}$	$\sum \lvert w_j\rvert^2 j^{2p} e^{2aj}$	same, exponent called s (p. 277)
two-sided $\ell_b^{a,s}$	weight $\langle l\rangle^{2s} e^{2a\lvert l\rvert}$	same (p. 279)
u, u_t	$U_m(q)$, $V_m(p)$	same, $\mathcal{S}q$, $\mathcal{S}'p$ (p. 277)
H	$\frac{1}{2}\sum \lambda_j(q_j^2+p_j^2) + \frac{1}{4}\int U^4$	same, (8)–(9) p. 280
quartic coefficient	$g_{ijkl} = \int_0^\pi \phi_i\phi_j\phi_k\phi_l$ (unweighted)	$G_{ijkl} = g_{ijkl}/\sqrt{\lambda_i\lambda_j\lambda_k\lambda_l}$, (10) p. 281; (11) correct
symplectic form	$\sum dq \wedge dp = i\sum dz \wedge d\bar{z}$	same (pp. 278, 281)
z	$(q+ip)/\sqrt{2}$	same (p. 281)
recovery of q, p	$q = \sqrt{2}\,\mathrm{Re}\, z$, $p = \sqrt{2}\,\mathrm{Im}\, z$	$q = \mathrm{Re}\, z$, $p = \mathrm{Im}\, z$ (p. 290): **disagrees**
action	$I_j = \lvert z_j\rvert^2 = (q_j^2+p_j^2)/2$	$I_j = q_j^2+p_j^2$ (p. 276, informal); $I = \frac{1}{2}\lvert z\rvert^2$ (p. 287); $\Lambda = \langle\alpha, I\rangle$ with $\alpha = \lambda$ (p. 290) requires $I = \lvert z\rvert^2$: **inconsistent**
quartic in z	$\frac{1}{16}\sum g_{ijkl}\prod(z+\bar{z})/\sqrt{\lambda\lambda\lambda\lambda}$	$\frac{1}{4}\sum G_{ijkl}\prod(z+\bar{z})$, then $\sum' G_{ijkl}w^4$ (p. 282): **two factor-4 slips**
normal form	$\frac{1}{2}\langle AI, I\rangle + \langle BI, Z\rangle$, $A_{ii} = \frac{9}{8\pi\lambda_i^2}$, $A_{ij} = B_{ji} = \frac{3}{2\pi\lambda_i\lambda_j}$	$\bar{G}_{ij} = \frac{6}{\pi}\frac{4-\delta_{ij}}{\lambda_i\lambda_j}$ (pp. 282, 284), $A = B = \bar{G}$ (p. 290): 16× ours
rotation sign	$\dot{z}_j = -i\lambda_j z_j$, φ increases	$\dot{z} = i\omega z$, flow $e^{i\omega t}$ (pp. 287, 289–290): time-reversed
torus angles	period one, $\nu = \omega/2\pi$	period 2π

Remark 1.14 (Why the disagreements are harmless for existence but fatal for constants). Every nondegeneracy condition used by the source (P96-W (A_1)–(A_3), p. 288) is invariant under multiplying (A, B) by a common positive constant and under rescaling I, and H is even in p, so the time reversal $(q,p)(t) \mapsto (q,-p)(-t)$ maps solutions to solutions. Hence the qualitative theorem is unaffected. The printed constants, frequency shifts, and the amplitude–action dictionary are not, and must not be copied.

2. Milestone W0: cubic attachment and quartic normalization

Throughout, Standing assumption 1.1 ($m > 0$) and the conventions of §1 are in force. The results of this part are: (a) the coefficient system (2) is equivalent to the PDE (Proposition 2.6); (b) $G_q : \ell^{1,1} \to \ell^{1,2}$ is a bounded cubic polynomial with an explicit constant (Theorem 2.5); (c) closed form of the quartic coefficients (Lemma 2.2); (d) the action coefficients A, B of the partial quartic Birkhoff normal form for $J = \{1, 2\}$ (Theorem 2.10); (e) the nondegeneracy conditions of the source for every $m > 0$ (Theorem 2.13); (f) the Duffing consistency check (Proposition 2.15). All proofs are complete except where a step is explicitly cited to a pinned source and labelled as such; those citations belong to later milestones (W1, W3) and are listed at the end (§2.9).

2.1. **Reconstruction.**

Lemma 2.1 (Reconstruction is bounded into C^k)**.** *Let $a > 0$, $p \in \mathbb{R}$, $k \in \mathbb{N}$. For $c \in \ell^{a,p}_{\mathbb{C}}$,*

$$\sum_{j\ge 1} |c_j|\, j^k \le M_{a,p,k}\|c\|_{a,p}, \qquad M_{a,p,k}^2 = \sum_{j\ge 1} j^{2k-2p} e^{-2aj} < \infty.$$

Consequently the series $\sum_j c_j\phi_j$ and all its termwise x-derivatives of order $\le k$ converge absolutely and uniformly on $[0,\pi]$; $c \mapsto \sum_j c_j\phi_j$ is a bounded linear map $\ell^{a,p} \to C^k([0,\pi])$ of norm $\le \sqrt{2/\pi}\, M_{a,p,k}$, its value vanishes at $x = 0, \pi$, and $\int_0^\pi (\sum_j c_j\phi_j)\phi_l\, dx = c_l$. In particular U_m, V_m and $q \mapsto U_m(\lambda q)$ are bounded linear maps $\ell^{a,p} \to C^k([0,\pi])$ for every $p \in \mathbb{R}$, because $\lambda_j^{\pm 1/2} \le (1+m)^{1/4} j^{1/2}$ and multiplication by (λ_j^r) maps $\ell^{a,p}$ into $\ell^{a,p-|r|}$ for $|r| \le 2$, $r \ge 0$, with norm $\le (1+m)^{r/2}$.

Proof. Cauchy–Schwarz: $\sum |c_j| j^k = \sum (|c_j| j^p e^{aj})(j^{k-p}e^{-aj}) \le \|c\|_{a,p} M_{a,p,k}$, and $M_{a,p,k} < \infty$ because $a > 0$. Since $|\partial_x^r \phi_j| \le \sqrt{2/\pi}\, j^r$, the Weierstrass M-test gives uniform convergence of the derivative series of order $r \le k$, hence termwise differentiation. Boundary values and the coefficient formula follow termwise from $\phi_j(0) = \phi_j(\pi) = 0$ and orthonormality. The bound on λ_j^r is $j^2 + m \le (1+m) j^2$. □

Source: P96-W Lemma 1, p. 278, first two sentences (asserted there without the estimate).

2.2. **The quartic coefficients.**

Lemma 2.2 (Closed form, (c))**.** *For $i, j, k, l \ge 1$ let $g_{ijkl} = \int_0^\pi \phi_i\phi_j\phi_k\phi_l\, dx$. Then*

$$g_{ijkl} = \frac{1}{2\pi}\Big([i+l=j+k] + [i+k=j+l] + [i+j=k+l] \\ - [i=j+k+l] - [j=i+k+l] - [k=i+j+l] - [l=i+j+k] \Big),$$

where $[\cdot]$ is the Iverson bracket. In particular g is symmetric, $g_{ijkl} = 0$ unless $\pm i \pm j \pm k \pm l = 0$ for some choice of signs, $g_{iiii} = \frac{3}{2\pi}$, and $g_{iijj} = \frac{1}{\pi}$ for $i \ne j$. Equivalently, for $l \ge 1$ and $i, j, k \ge 1$,

$$g_{ijkl} = \frac{1}{2\pi} \sum_{\varepsilon \in \{\pm 1\}^3} s(\varepsilon)\, [\, l = \varepsilon_1 i + \varepsilon_2 j + \varepsilon_3 k\,], \qquad s(\varepsilon) = (-1)^{1+\#\{t:\varepsilon_t = -1\}}, \tag{3}$$

so that $|g_{ijkl}| \le \frac{1}{2\pi} \sum_\varepsilon [\, l = \varepsilon_1 i + \varepsilon_2 j + \varepsilon_3 k\,]$.

Proof. **Step S2.1: Product to sum.** $\sin\alpha \sin\beta = \frac{1}{2}(\cos(\alpha-\beta) - \cos(\alpha+\beta))$ gives

$$\phi_i\phi_j\phi_k\phi_l = \frac{1}{\pi^2}\big(\cos(i-j)x - \cos(i+j)x\big)\big(\cos(k-l)x - \cos(k+l)x\big).$$

Step S2.2: Cosine integrals. For integers A, B: $\int_0^\pi \cos Ax \cos Bx\, dx = \frac{\pi}{2}\big([A=B] + [A=-B]\big)$ (this includes $A = B = 0$, value π). Hence

$$g_{ijkl} = \frac{1}{2\pi}\Big([i-j=k-l] + [i-j=l-k] - [i-j=k+l] - [i-j=-k-l] \\ - [i+j=k-l] - [i+j=l-k] + [i+j=k+l] + [i+j=-k-l] \Big).$$

The last bracket vanishes for positive indices; rewriting the others gives the stated form. The special values are read off: for (i,i,i,i) the three positive brackets hold and no negative one; for (i,i,j,j), $i \ne j$, exactly $[i+l=j+k]$ and $[i+k=j+l]$ hold (a negative bracket would force one index to be the sum of the other three, e.g. $i = i + 2j$, impossible).

Step S2.3: Signed form. The seven relations, solved for l, are $l=-i+j+k$, $l=i-j+k$, $l=i+j-k$ (sign $+$, one minus), $l=i-j-k$, $l=-i+j-k$, $l=-i-j+k$ (sign $-$, two minuses), $l=i+j+k$ (sign $-$, no minus); the eighth pattern $l=-i-j-k$ is impossible for $l\geq 1$ and may be added with either sign. This is (3). □

Source: P96-W (10)–(11), p. 281 (only the selection rule and G_{iijj} are stated, "by an elementary calculation – see [6]"); KP96 p. 13 (computation of G_{ijij}). The weighted coefficient of P96-W is $G_{ijkl}=g_{ijkl}/\sqrt{\lambda_i\lambda_j\lambda_k\lambda_l}$; its (11) agrees with the lemma.

2.3. **The cubic gradient.**

Lemma 2.3 (G and its gradient as absolutely convergent sums)**.** *Let $a>0$, $p\geq 0$, $q\in\ell^{a,p}$ and write $\hat q_j=q_j/\sqrt{\lambda_j}$. Then*

$$G(q)=\frac14\sum_{i,j,k,l\geq 1} g_{ijkl}\hat q_i\hat q_j\hat q_k\hat q_l,\qquad G_q(q)_l=\frac{1}{\sqrt{\lambda_l}}\sum_{i,j,k\geq 1} g_{ijkl}\hat q_i\hat q_j\hat q_k,$$

both sums converging absolutely; G is Gâteaux differentiable along every coordinate with $\partial G/\partial q_l=G_q(q)_l$; and $\int_0^\pi U_m(q)^3\phi_l\,dx=\sqrt{\lambda_l}\,G_q(q)_l$.

Proof. By Lemma 2.1 with $k=0$, $S=\sum_j|\hat q_j|<\infty$ and $U_m(q)=\sum\hat q_j\phi_j$ uniformly. Since $|\phi_j|\leq\sqrt{2/\pi}$, the quadruple series $\sum\hat q_i\hat q_j\hat q_k\hat q_l\phi_i\phi_j\phi_k\phi_l$ is dominated by $(2/\pi)^2S^4$ uniformly in x; integrate termwise (dominated convergence for series) to obtain both formulas and their absolute convergence. Differentiating $\frac14\int U_m(q+te_l)^4$ at $t=0$ under the integral (the integrand is a polynomial in t with coefficients continuous in x) gives $\int U_m(q)^3\phi_l/\sqrt{\lambda_l}$, which is $G_q(q)_l$ by Definition 1.9 and the triple-series formula. □

Lemma 2.4 (Weighted convolution algebra)**.** *Let $a\geq 0$ and $s\geq 1$. For $c,d\in\ell^{a,s}_b$,*

$$\|c*d\|_{a,s}\leq K_s\|c\|_{a,s}\|d\|_{a,s},\qquad K_s=2^s\Big(1+2\zeta(2s)\Big)^{1/2}.$$

In particular $K_{3/2}^2=8(1+2\zeta(3))$. The convolution sum converges absolutely for each l.

Proof. Let $w_l=\langle l\rangle^s e^{a|l|}$, $C_l=w_l|c_l|$, $D_l=w_l|d_l|$.

Step S2.4: Weight inequality. For $i,l\in\mathbb{Z}$: $e^{a|l|}\leq e^{a|i|}e^{a|l-i|}$ and $\langle l\rangle\leq\langle i\rangle+\langle l-i\rangle$ (if $|l|\leq 1$ the left side is 1; otherwise $|l|\leq|i|+|l-i|$). By convexity of $t\mapsto t^s$, $(\alpha+\beta)^s\leq 2^{s-1}(\alpha^s+\beta^s)$, so $w_l\leq 2^{s-1}(w_ie^{a|l-i|}+e^{a|i|}w_{l-i})\leq 2^{s-1}(w_iw_{l-i}/\langle l-i\rangle^s+w_iw_{l-i}/\langle i\rangle^s)$.

Step S2.5: Young and Cauchy–Schwarz. Hence $w_l|(c*d)_l|\leq 2^{s-1}\sum_i\big(C_i\,|d_{l-i}|e^{a|l-i|}+|c_i|e^{a|i|}D_{l-i}\big)=2^{s-1}\big((C*D')_l+(C'*D)_l\big)$ with $C_i'=C_i/\langle i\rangle^s$, $D_i'=D_i/\langle i\rangle^s$. Young's inequality $\|X*Y\|_{\ell^2}\leq\|X\|_{\ell^2}\|Y\|_{\ell^1}$ and $\|D'\|_{\ell^1}\leq\|D\|_{\ell^2}(\sum_i\langle i\rangle^{-2s})^{1/2}=\|D\|_{\ell^2}(1+2\zeta(2s))^{1/2}$ give the claim; the same bounds give absolute convergence of each convolution sum. □

Source: P96-W Lemma 2, p. 279 and Appendix A, pp. 294–295; KP96 Lemma 2 p. 12 and Appendix A, pp. 31–32 (constant "depending only on s", not made explicit there). This lemma is the W1 node `bpW:w1:algebra`; it is proved here because (b) needs it.

Theorem 2.5 (Cubic gradient, (b))**.** *Let $a\geq 0$, $p\geq\frac12$, and define the symmetric trilinear map T on $(\ell^{a,p}_{\mathbb{C}})^3$ by*

$$T(q,q',q'')_l=\frac{1}{\sqrt{\lambda_l}}\sum_{i,j,k\geq 1} g_{ijkl}\,\hat q_i\hat q_j'\hat q_k''\qquad(\hat q_j=q_j/\sqrt{\lambda_j}).$$

The sums converge absolutely, T is a bounded trilinear map into $\ell^{a,p+1}_{\mathbb{C}}$, real on real arguments, and

$$\|T(q,q',q'')\|_{a,p+1}\leq C_*\|q\|_{a,p}\|q'\|_{a,p}\|q''\|_{a,p},\qquad C_*=\frac{K_{p+1/2}^2}{\pi}.$$

For $a > 0$, $G_q(q) = T(q,q,q)$ (Lemma 2.3). Consequently, for $a = p = 1$, $G_q : \ell^{1,1} \to \ell^{1,2}$ is real analytic (indeed a continuous cubic polynomial), and

$$\|G_q(q)\|_{1,2} \le C_m \|q\|_{1,1}^3, \qquad C_m = \frac{8(1+2\zeta(3))}{\pi} \approx 8.67.$$

The constant is independent of $m > 0$ (we only use $\lambda_j \ge j$).

Proof. **Step S2.6: Even majorants.** Given $q \in \ell^{a,p}_{\mathbb{C}}$ define the even two-sided sequence $x_i = |q_{|i|}|/\sqrt{\lambda_{|i|}}$ for $i \neq 0$, $x_0 = 0$; similarly x', x''. Using $\lambda_i \ge i$,

$$\|x\|^2_{a,p+1/2} = 2\sum_{i\ge1} \frac{i}{\lambda_i}\, i^{2p} e^{2ai} |q_i|^2 \le 2\|q\|^2_{a,p}.$$

Step S2.7: Selection rule bound. By (3), for $l \ge 1$,

$$\sum_{i,j,k\ge1} |g_{ijkl}|\, |\hat q_i \hat q'_j \hat q''_k| \le \frac{1}{2\pi} \sum_{i,j,k\ge1} \sum_{\varepsilon} [\, l = \varepsilon_1 i + \varepsilon_2 j + \varepsilon_3 k\,]\, x_i x'_j x''_k \le \frac{1}{2\pi}(x * x' * x'')_l,$$

since $(i,j,k,\varepsilon) \mapsto (\varepsilon_1 i, \varepsilon_2 j, \varepsilon_3 k)$ is injective into $(\mathbb{Z}\setminus 0)^3$ and x is even. Thus the sum defining T converges absolutely.

Step S2.8: Smoothing. With $X = x * x' * x''$ (even, by evenness of the factors) and $l/\lambda_l \le 1$,

$$\|T\|^2_{a,p+1} \le \sum_{l\ge1} \frac{l^{2p+2} e^{2al}}{\lambda_l} \frac{X_l^2}{4\pi^2} \le \frac{1}{4\pi^2} \sum_{l\ge1} l^{2p+1} e^{2al} X_l^2 \le \frac{1}{8\pi^2} \|X\|^2_{a,p+1/2}.$$

By Lemma 2.4 with $s = p + \frac12 \ge 1$, $\|X\|_{a,p+1/2} \le K_s^2 \|x\|\|x'\|\|x''\| \le K_s^2 2^{3/2} \|q\|\|q'\|\|q''\|$. Hence $\|T\|_{a,p+1} \le (2\sqrt{2}\pi)^{-1} 2^{3/2} K_s^2 \prod\|\cdot\| = (K_s^2/\pi)\prod\|\cdot\|$.

Step S2.9: Analyticity. Trilinearity and symmetry are clear from the formula and the symmetry of g. A bounded symmetric trilinear map T between complex Banach spaces defines the continuous homogeneous cubic polynomial $q \mapsto T(q,q,q)$, which is analytic on the whole space, with the finite power series $T(q_0+h)^3$ expanded by multilinearity (in Lean: a `ContinuousMultilinearMap` is analytic). It maps real sequences to real sequences. For $a > 0$ the identification with G_q is Lemma 2.3. □

Source: P96-W Lemma 3, pp. 279–280 (with $O(\cdot)$ instead of a constant) and (13)–(14), p. 284; KP96 p. 16. The source deduces analyticity from coordinatewise analyticity and local boundedness [P96-W ref. 9, Appendix A]; the polynomial structure makes that principle unnecessary here.
SOURCE_GAP: P96-W p. 280 asserts $\|\mathcal{S}q\|_{a,s+1/2} \le \|q\|_{a,s}$ in the two-sided norm. The correct bound is $\le \sqrt{2}\|q\|_{a,s}$ (Step S2.6); the operator norm is exactly $\sqrt{2}$ for every fixed $m > 0$ (take $q = e_j$, $j \to \infty$), so the printed bound fails for every $m > 0$ (e.g. $q = e_1$, $a = s = 0$, $m = 1$). Constants only; recorded as new item N-1 in the errata check.

2.4. PDE ↔ coefficient system.

Proposition 2.6 (Equivalence, (a)). *Let $I \subset \mathbb{R}$ be an open interval and $q, p : I \to \ell^{1,1}$.*

- *(i) (Coefficients ⇒ PDE.) Suppose q, p are continuous into $\ell^{1,1}$, continuously differentiable into $\ell^{1,0}$, and satisfy (2) as an identity in $\ell^{1,0}$. Then $u(t,x) = U_m(q(t))(x)$ is a classical solution of (1) on I (Definition 1.2) with $u_t(t,\cdot) = V_m(p(t))$.*
- *(ii) (PDE ⇒ coefficients.) Suppose $u(t,x) = U_m(q(t))(x)$ is a classical solution of (1) on I and $u_t(t,\cdot) = V_m(p(t))$ for every t. Then every q_j, p_j is C^1 on I and (2) holds coordinatewise.*

In both directions the cubic term is realized exactly: for $q \in \ell^{1,1}$,

$$U_m(q)^3 = \sum_{l\geq 1} \sqrt{\lambda_l}\, G_q(q)_l\, \phi_l \quad \text{uniformly on } [0,\pi], \qquad -\partial_x^2 U_m(q) + mU_m(q) = U_m(\lambda^2 q), \tag{4}$$

where $(\lambda^2 q)_j = \lambda_j^2 q_j$ and the right sides converge in C^0 (first) and C^0 (second, as $\lambda^2 q \in \ell^{1,-1}$).

Proof. **Step S2.10: Force identity.** Let $f = U_m(q)^3 \in C([0,\pi])$. By Lemma 2.3, $\int f\phi_l = \sqrt{\lambda_l} G_q(q)_l$; by Theorem 2.5 and Lemma 2.1 (the sequence $(\sqrt{\lambda_l} G_q(q)_l)$ lies in $\ell^{1,1}$), $S = \sum_l \sqrt{\lambda_l} G_q(q)_l \phi_l$ converges uniformly to a continuous S with the same sine coefficients as f. Since $(\phi_l)_{l\geq 1}$ is a complete orthonormal system of $L^2(0,\pi)$, $f = S$ almost everywhere, hence everywhere by continuity. The second identity is termwise differentiation (Lemma 2.1 with $k=2$) and $-\phi_j'' + m\phi_j = \lambda_j^2 \phi_j$.

Step S2.11: Time regularity. In (i), $\dot q = \lambda p$ is continuous into $\ell^{1,0}$, and since p is C^1 into $\ell^{1,0}$ and multiplication by λ is bounded $\ell^{1,0} \to \ell^{1,-1}$, $\dot q$ is C^1 into $\ell^{1,-1}$ with $\ddot q = \lambda \dot p = -\lambda^2 q - \lambda G_q(q)$. As $U_m : \ell^{1,-1} \to C^2([0,\pi])$ is bounded linear, $t \mapsto u(t,\cdot)$ is C^2 into $C^2([0,\pi])$; therefore $u, u_t, u_{tt}, u_x, u_{xx}$ exist and are jointly continuous, $u_t = U_m(\lambda p) = V_m(p)$ and $u_{tt} = U_m(\ddot q)$.

Step S2.12: Identification. By (4), $U_m(\lambda^2 q) = -u_{xx} + mu$ and $U_m(\lambda G_q(q)) = \sum_l \sqrt{\lambda_l} G_q(q)_l \phi_l = u^3$; hence $u_{tt} = u_{xx} - mu - u^3$. The boundary conditions hold by Lemma 2.1.

Step S2.13: Converse. In (ii), $\int_0^\pi u(t,x)\phi_l(x)\,dx = q_l(t)/\sqrt{\lambda_l}$ and $\int_0^\pi u_t \phi_l = \sqrt{\lambda_l} p_l$. Differentiation under the integral sign (u_t, u_{tt} continuous on compact sets) gives $\dot q_l = \sqrt{\lambda_l} \int u_t \phi_l = \lambda_l p_l$ and $\sqrt{\lambda_l}\dot p_l = \int u_{tt}\phi_l$. Two integrations by parts, using $u(t,0) = u(t,\pi) = 0 = \phi_l(0) = \phi_l(\pi)$, give $\int u_{xx}\phi_l = -l^2 \int u\phi_l$, so $\int u_{tt}\phi_l = -\lambda_l^2 q_l/\sqrt{\lambda_l} - \sqrt{\lambda_l} G_q(q)_l$ by Lemma 2.3. Divide by $\sqrt{\lambda_l}$. □

Source: P96-W Lemma 1, p. 278, and (7), p. 278.
Expansion/strengthening of the source proof: P96-W Lemma 1 (p. 278) argues via convergence in complex neighbourhoods, termwise time differentiation and completeness, for real-analytic curves. The proof above expands these steps and proves a stronger C^1-curve version (i); this is not a source gap. (Separately, the second line of the source's u_{tt} display drops a minus sign on $\lambda_j^2 q_j$: errata item N-4; the proof above uses the correct sign.)
completeness of (ϕ_l) in $L^2(0,\pi)$ is a standard external fact (Fourier basis of L^2 on the circle applied to odd extensions). The all-time real-analytic version used by the headline is W2.

2.5. **Quartic nonresonance.**

Lemma 2.7 (Qualitative nonresonance of all quartic terms)**.** *Let $m > 0$, let $a,b,c,d \geq 1$ with $g_{abcd} \neq 0$, and let $\sigma \in \{\pm 1\}^4$. If $\sigma_1\lambda_a + \sigma_2\lambda_b + \sigma_3\lambda_c + \sigma_4\lambda_d = 0$, then the indices with $\sigma = +1$ and those with $\sigma = -1$ form equal multisets (in particular there are two of each).*

Proof. By Lemma 2.2, $g_{abcd} \neq 0$ implies a signed relation $\pm a \pm b \pm c \pm d = 0$; in particular each index is at most the sum of the other three.

Step S2.14: Four or three equal signs. All signs equal is impossible ($\lambda > 0$). If three signs agree, say $\lambda_a + \lambda_b + \lambda_c = \lambda_d$, then $d \leq a+b+c$, so $\lambda_d \leq \lambda_{a+b+c}$, while $\lambda_x + \lambda_y > \lambda_{x+y}$ for $x, y \geq 1$ (square both sides: $2\sqrt{(x^2+m)(y^2+m)} > 2xy - m$); hence $\lambda_a + \lambda_b + \lambda_c > \lambda_{a+b+c} \geq \lambda_d$.

Step S2.15: Two against two. Let $\lambda_a + \lambda_b = \lambda_c + \lambda_d$ with $\{a,b\} \neq \{c,d\}$ as multisets, and let $f(s) = \sqrt{s^2+m}$. Up to renaming within and between the two sides, the signed relation is one of: (1) $a+b=c+d$: for fixed sum, $f(x)+f(y)$ is strictly increasing in $|x-y|$ by strict convexity of f, so equality forces $|a-b| = |c-d|$ and $\{a,b\} = \{c,d\}$; (2) $a-b = d-c =: t$ (that is, $a+c = b+d$): then $f(b+t)+f(b) = f(c)+f(c+t)$ and $y \mapsto f(y)+f(y+t)$ is strictly increasing on $\{y \geq 1,\ y+t \geq 1\}$, so $b=c$, $a=d$; (3) one index is the sum of the other three, and after renaming it is $a = b+c+d$ (the side containing the largest index is called the left side). Then $\lambda_a - \lambda_c = \int_c^{b+c+d} f'$ and $\lambda_d - \lambda_b = \int_b^d f'$

(nonpositive if $d \le b$). Since $f' > 0$ is increasing, $\int_c^{c+b+d} f' > \int_{c+2b}^{c+b+d} f' \ge \int_b^d f'$ when $d > b$ (an interval of the same length $d-b$ shifted right by $c+b>0$). So $\lambda_a+\lambda_b > \lambda_c+\lambda_d$. Every signed relation among four positive integers with two signs on each side is of type (1) or (2) after renaming, and every relation with one sign different is of type (3); this exhausts the cases. □

Remark 2.8. The lemma is qualitative and covers *all* indices. It does not give a uniform lower bound: divisors such as $\lambda_{j+1}-\lambda_j-\lambda_{k+1}+\lambda_k$ tend to 0, which is why P96-W (p. 276) says no complete quartic normal form is available. The quantitative bound for quartic terms with at least one index in $\{1,\dots,n\}$, $|\lambda_i'+\lambda_j'+\lambda_k'+\lambda_l'| \ge cm/\sqrt{n^2+m}^3$, is P96-W Lemma 4, p. 283 (proof pp. 285–286); it belongs to W3 and is cited, not reproved, here.

2.6. **Partial quartic Birkhoff normal form: the action coefficients.** Write $\hat\lambda = (\lambda_j)$ and, in the complex coordinates of Definition 1.11, $q_j = (z_j+\bar z_j)/\sqrt{2}$. Then

$$G = \frac{1}{4}\sum_{i,j,k,l} \frac{g_{ijkl}}{\sqrt{\lambda_i\lambda_j\lambda_k\lambda_l}} q_iq_jq_kq_l = \frac{1}{16}\sum_{i,j,k,l} \frac{g_{ijkl}}{\sqrt{\lambda_i\lambda_j\lambda_k\lambda_l}} (z_i+\bar z_i)(z_j+\bar z_j)(z_k+\bar z_k)(z_l+\bar z_l). \tag{5}$$

Expanding gives a sum of monomials $z^\alpha \bar z^\beta$ with $|\alpha|+|\beta|=4$. Along the linear flow ($\dot z_j = -i\lambda_j z_j$) such a monomial evolves by $e^{-i\langle \alpha-\beta,\hat\lambda\rangle t}$; it is *resonant* if $\langle \alpha-\beta,\hat\lambda\rangle = 0$. Let $J=\{1,2\}$, $N=\{j\ge 3\}$, and split $G = G^{\mathrm{tan}} + G^{\mathrm{nor}}$, where G^{nor} collects the monomials all of whose indices lie in N.

Definition 2.9 (Partial quartic normal form)**.** $\bar G$ is the sum of the resonant monomials of G^{tan} (with their coefficients), and the partial quartic normal form is $\Lambda + \bar G + G^{\mathrm{nor}}$.

Theorem 2.10 (Action coefficients, (d))**.** *For $J=\{1,2\}$ and every $m>0$, with $I=(I_1,I_2)$, $Z=(I_j)_{j\ge3}$ and $I_j=|z_j|^2$,*

$$\bar G = \frac{1}{2}\langle AI, I\rangle + \langle BI, Z\rangle, \qquad A_{ii} = \frac{9}{8\pi\lambda_i^2}, \quad A_{12}=A_{21} = \frac{3}{2\pi\lambda_1\lambda_2}, \quad B_{ji} = \frac{3}{2\pi\lambda_j\lambda_i}\ (j\ge3,\ i\in J).$$

Equivalently $A = \frac{3}{8\pi} D(4\mathbf{1}\mathbf{1}^{\mathsf T} - \mathrm{Id})D$ with $D=\mathrm{diag}(\lambda_1^{-1},\lambda_2^{-1})$, and $(BI)_j = \frac{3}{2\pi\lambda_j}\big(\frac{I_1}{\lambda_1}+\frac{I_2}{\lambda_2}\big)$. More generally, the full angle average of G is

$$\langle G\rangle = \frac{3}{16\pi}\Big(4\Big(\sum_{j\ge1}\frac{I_j}{\lambda_j}\Big)^2 - \sum_{j\ge1}\frac{I_j^2}{\lambda_j^2}\Big). \tag{6}$$

Proof. **Step S2.16: Resonant equals averaged.** By Lemma 2.7, a monomial $z^\alpha\bar z^\beta$ of (5) with nonzero coefficient is resonant iff $\alpha=\beta$, i.e. it is $|z_a|^2|z_b|^2 = I_aI_b$. With $z_j = \sqrt{I_j}e^{-i\varphi_j}$ the average over independent angles of $z^\alpha\bar z^\beta$ is $\delta_{\alpha\beta}I^\alpha$. Hence $\bar G$ is the part of $\langle G\rangle$ (average over all angles, computed on finitely supported q and extended by the absolutely convergent formula) consisting of the terms that contain at least one tangential action.

Step S2.17: Averages of monomials. With $q_j = \sqrt{2I_j}\cos\varphi_j$: $\langle q_iq_jq_kq_l\rangle = 0$ unless the indices pair up; $\langle q_i^4\rangle = 4I_i^2\cdot\frac38 = \frac32 I_i^2$ and $\langle q_i^2q_j^2\rangle = I_iI_j$ for $i\ne j$. Among ordered quadruples, (i,i,i,i) occurs once and the multiset $\{i,i,j,j\}$ $(i<j)$ six times.

Step S2.18: Summation. Therefore, with $y_j = I_j/\lambda_j$ and Lemma 2.2,

$$\langle G\rangle = \frac14\Big(\sum_i \frac{3}{2\pi}\cdot\frac32 y_i^2 + \sum_{i<j} 6\cdot\frac{1}{\pi}\,y_iy_j\Big) = \frac{9}{16\pi}\sum_i y_i^2 + \frac{3}{2\pi}\sum_{i<j} y_iy_j,$$

which is (6). Matching with $\frac12\sum_{i,j}A_{ij}I_iI_j = \frac12\sum_i A_{ii}I_i^2 + \sum_{i<j}A_{ij}I_iI_j$ on J and with $\sum_{j\ge3,i\in J}B_{ji}I_iZ_j$ on the mixed terms gives the coefficients. □

Proposition 2.11 (Formal partial Birkhoff transformation)**.** *Let F be the quartic polynomial whose coefficient on each nonresonant monomial $z^\alpha \bar{z}^\beta$ of $G^{\tan}$ with coefficient $c_{\alpha\beta}$ is $c_{\alpha\beta}/(-i\langle \alpha-\beta, \hat{\lambda}\rangle)$, and zero otherwise. Then F is real ($\overline{F} = F$ under $z \leftrightarrow \bar{z}$), $X_\Lambda F = G^{\tan} - \bar{G}$ where $X_\Lambda F = \frac{d}{dt} F$ along the linear flow, and formally*

$$H \circ \Phi_F^1 = \Lambda + \bar{G} + G^{\mathrm{nor}} + (\textit{terms of degree } \geq 6),$$

where Φ_F^1 is the time-one map of X_F. The quartic part of the normal form is independent of the choice of F among solutions of $X_\Lambda F = G^{\tan} - \bar{G}$.

Proof. The divisors are nonzero by Lemma 2.7; $X_\Lambda(z^\alpha \bar{z}^\beta) = -i\langle \alpha-\beta, \hat{\lambda}\rangle z^\alpha \bar{z}^\beta$ gives the homological identity. Reality: the coefficient of $\bar{z}^\alpha z^\beta$ is $\bar{c}_{\alpha\beta}$ (since G is real) and the divisor changes sign and is conjugated. The Lie series $H \circ \Phi_F^1 = H + \{H,F\} + \frac{1}{2}\{\{H,F\},F\} + \dots$ with $\{H,F\} = X_F H = -X_H F$ has quartic part $G - X_\Lambda F = G^{\mathrm{nor}} + \bar{G}$. Any two solutions F differ by a resonant quartic F_0, and $X_\Lambda F_0 = 0$. □

Source: P96-W Main Proposition, pp. 281–282; proof pp. 282–284, (12) and the Taylor formula on p. 283. The analytic realization of Φ_F^1 on a neighbourhood of 0 in $E_{a,p}$, the smoothing of X_F, and the bound $O(\|z\|^6)$ on the remainder are W3 (they use P96-W Lemma 4 and (13)–(14)). The *coefficients* printed there, $\bar{G}_{ij} = \frac{6}{\pi}\frac{4-\delta_{ij}}{\lambda_i \lambda_j}$, are 16 times those of Theorem 2.10; see Remark 2.12.

Remark 2.12 (Comparison with the printed coefficients). P96-W p. 282 expands G as $\frac{1}{4}\sum G_{ijkl}(z_i + \bar{z}_i)\cdots(z_l + \bar{z}_l)$; by (5) the prefactor is $\frac{1}{16}$. The next line rewrites this as $\sum' G_{ijkl} w_i w_j w_k w_l$ *without* any prefactor, and the multiplicity count of p. 284 ($\bar{G}_{ij} = 24 G_{iijj}$, $\bar{G}_{ii} = 12 G_{iiii}$) is correct for that line. Two factors of 4 are therefore lost; with the prefactor $\frac{1}{16}$ the same count gives exactly A and B above. The factor is common to A and B, so the nondegeneracy arguments of P96-W Lemmas 5–6 are unaffected (Remark 1.14).

2.7. **Nondegeneracy for every $m > 0$.** The Cantor Manifold Theorem used by P96-W (p. 288, assumption A) requires, with $\omega(I) = \lambda_J + AI$ and $\Omega(I) = \lambda_N + BI$, ($\mathrm{A}_1$) $\det A \neq 0$; (A_2) $\langle l, \lambda_N\rangle \neq 0$; ($\mathrm{A}_3$) $\langle k, \omega(I)\rangle + \langle l, \Omega(I)\rangle \not\equiv 0$ as an affine function of I, for all $(k,l) \in \mathbb{Z}^2 \times \mathbb{Z}^N$ with $1 \leq |l| \leq 2$ ($|l| = \sum |l_j|$).

Theorem 2.13 (Nondegeneracy, (e))**.** *For $J = \{1,2\}$ and every $m > 0$:*

(i) $\det A = -\dfrac{63}{64\pi^2 \lambda_1^2 \lambda_2^2} = -\dfrac{63}{64\pi^2 (1+m)(4+m)} \neq 0$*;*
(ii) $\langle l, \lambda_N\rangle \neq 0$ *for* $1 \leq |l| \leq 2$*;*
(iii) (A_3) *holds.*

Proof. **Step S2.19: Determinant.** $\det A = (\frac{3}{8\pi})^2 \lambda_1^{-2}\lambda_2^{-2} \det\left(\begin{smallmatrix} 3 & 4 \\ 4 & 3 \end{smallmatrix}\right) = -\frac{63}{64\pi^2}\lambda_1^{-2}\lambda_2^{-2}$. (For general n: $\det(4\mathbf{1}\mathbf{1}^{\mathsf{T}} - \mathrm{Id}) = (-1)^{n-1}(4n-1)$.)

Step S2.20: Pure normal. $\langle l, \lambda_N\rangle$ is $\pm\lambda_a$, $\pm 2\lambda_a$, $\pm(\lambda_a + \lambda_b)$ or $\pm(\lambda_a - \lambda_b)$ with $a \neq b$ in N; all are nonzero since $\lambda > 0$ is strictly increasing.

Step S2.21: Reduction of ($\mathbf{A}_3$). Let $b = (\lambda_1^{-1}, \lambda_2^{-1})$ and $s = \sum_{j \geq 3} l_j/\lambda_j$. Then $A = \frac{3}{8\pi}(4bb^{\mathsf{T}} - D^2)$ and $B^{\mathsf{T}} l = \frac{3}{2\pi} s\, b$, so the linear part of the affine map is $\frac{3}{8\pi}\big(4(\langle b, k\rangle + s)b - D^2 k\big)$. It vanishes iff $k_i/\lambda_i = c$ for $i = 1,2$ with $c = 4(\langle b, k\rangle + s)$; then $\langle b, k\rangle = 2c$ and $s = -\frac{7}{4}c$. The constant part is $c(\lambda_1^2 + \lambda_2^2) + \langle l, \lambda_N\rangle$.

Step S2.22: The case $|l| = 1$. $l = \sigma e_\nu$, $\nu \geq 3$: $s = \sigma/\lambda_\nu$, $c = -4\sigma/(7\lambda_\nu)$, and the constant part vanishes iff $7\lambda_\nu^2 = 4(\lambda_1^2 + \lambda_2^2)$. Then $k_i^2 = c^2\lambda_i^2 = \frac{4\lambda_i^2}{7(\lambda_1^2+\lambda_2^2)} \in (0, \frac{4}{7})$, not the square of an integer. So (A_3) holds.

Step S2.23: The case $|l| = 2$. $|s| \le 2/\lambda_3 < \frac{2}{3}$, so $|c| < \frac{8}{21}$, and $k_2 - k_1 = c(\lambda_2 - \lambda_1)$ with $0 < \lambda_2 - \lambda_1 < 1$; integrality forces $k_1 = k_2$, hence $c = 0$, hence $s = 0$. But $s \neq 0$ for $|l| = 2$: $s = 0$ would need $l = e_a - e_b$ with $\lambda_a = \lambda_b$, $a \neq b$. So the linear part does not vanish. □

Source: P96-W assumption A p. 288; Lemma 5 pp. 291–292 ((A_1), with the printed matrix; the printed $(1-4n)\prod \lambda_j^{-2}$ agrees after scaling); Lemma 6 pp. 292–293 ((A_3) for admissible J). Our proof of (iii) is the specialization of Lemma 6 to $J = \{1, 2\}$; the second branch of the source's $|l| = 2$ argument ($|\langle w, l\rangle| \ge (4n-1)/(4n-4)$) cannot occur for $J = \{1, 2\}$ since $|\langle w, l\rangle| \le 2/\lambda_3 < 1$.

Remark 2.14 (Spectral asymptotics with the corrected B). $\Omega_j(I) = \lambda_j + \frac{3}{2\pi\lambda_j}\langle b, I\rangle = j + \frac{m_I}{j} + O(j^{-3})$ with $m_I = \frac{m}{2} + \frac{3}{2\pi}\langle b, I\rangle$, so $(\Omega_i - \Omega_j)/(i-j) = 1 + O(j^{-2})$ uniformly for bounded I: the exponent $\kappa = 2$ of P96-W p. 291 is unchanged by the correction (only $v = \frac{24}{\pi}b$ becomes $\frac{3}{2\pi}b$).

2.8. Duffing consistency check.

Proposition 2.15 (One-mode frequency shift, (f))**.** *Restrict (1) to $u = x(t)\phi_j(x)$ with only mode j excited and drop the forcing of other modes (which is $O(x^3)$ and does not enter the first-order frequency shift). The coefficient equation is the Duffing equation $\ddot{x} + \lambda_j^2 x + \varepsilon x^3 = 0$ with $\varepsilon = g_{jjjj} = \frac{3}{2\pi}$, whose Lindstedt–Poincaré frequency at amplitude a is $\lambda_j + \frac{3\varepsilon a^2}{8\lambda_j} + O(a^4)$. With $q_j = \sqrt{\lambda_j}x$ and $I_j = \frac{1}{2}\lambda_j a^2$ this is*

$$\lambda_j + \frac{9}{16\pi\lambda_j}a^2 = \lambda_j + \frac{9}{8\pi\lambda_j^2}I_j = \lambda_j + A_{jj}I_j,$$

the frequency $\partial_{I_j}(\lambda_j I_j + \frac{1}{2}A_{jj}I_j^2)$ of the normal form. The printed P96-W value $\bar{G}_{jj} = \frac{18}{\pi\lambda_j^2}$ would predict a shift 16 times larger.

Proof. $\int (x\phi_j)^3\phi_j = x^3 g_{jjjj}$ gives the Duffing coefficient. The Lindstedt shift $3\varepsilon a^2/(8\omega_0)$ is classical (e.g. by averaging εx^3 against cos over a period: $\frac{2}{T}\int \varepsilon a^3 \cos^4 = \frac{3}{4}\varepsilon a^3$). Along a solution $x = a\cos\varphi$, $q_j = \sqrt{\lambda_j}a\cos\varphi$ and $p_j = \dot{q}_j/\lambda_j$, so to leading order $I_j = \frac{1}{2}(q_j^2 + p_j^2) = \frac{1}{2}\lambda_j a^2$. Substitute. □

Remark 2.16. The relation between I_j and the amplitude holds to leading order only; the agreement checked is of the coefficient of the first-order shift, which is what the normal form determines. `scripts/wave/w0_checks.py` also integrates the Duffing equation numerically for $m \in \{0.1, 1, 10\}$ and confirms $|\omega_{\text{num}} - (\lambda_1 + A_{11}I_1)| = O(I_1^2)$.

2.9. Ledger: proved, cited, source gaps.

- *Proved here*: Lemmas 2.1, 2.2, 2.3, 2.4, 2.7; Theorems 2.5, 2.10, 2.13; Propositions 2.6, 2.11 (formal level), 2.15.
- *External standard facts*: completeness of (ϕ_l) in $L^2(0, \pi)$; analyticity of continuous multilinear maps; Young's inequality; $\zeta(3) < \infty$; the classical Lindstedt shift.
- *Cited, for later milestones*: quantitative divisor bound (P96-W Lemma 4, W3); analytic realization of the Birkhoff map and its remainder (P96-W Main Proposition, W3).
- *Expansions of source proofs*: P96-W Lemma 1 (expanded and strengthened to C^1 curves).
- *Source gaps / normalization defects*: P96-W p. 280 constant (N-1); P96-W p. 278 sign typo (N-4); KP96 p. 13 factor 2 (N-5); P96-W p. 282/284 factor 16 (E-PDE-W-1); P96-W pp. 287, 290 action/recovery/sign conventions (E-PDE-W-2).

3. The proposed headline W-NLW2

Status: PROPOSED (not frozen, not approved; for independent audit and owner review).

This section states the headline proposed in the scoping report (`docs/explorations/pde-kam-and-leapfrogging.md`, §2.3), adopted for the source/design phase by owner decision D30, with every definition written out in the conventions of §1. It is a proposal for later independent

statement audit and owner approval; nothing here is frozen. W-NLW2 is *the D30 first headline (a declared weakening of P96-W's Main Theorem for $J = \{1, 2\}$)*; it is not the full specialized Main Theorem. The weakenings are listed in Remark 3.7.

3.1. **Definitions.** Fix $m > 0$, the weights $a = p = 1$ (Convention 1.5), the phase space $E = E_{1,1}$ with norm $\|(q,p)\|_E = \max(\|q\|_{1,1}, \|p\|_{1,1})$, the reconstruction U_m, V_m (Definition 1.7) and the cubic gradient G_q (Definition 1.9). Let $P = (0,\infty)^2$, $r(I) = \sqrt{I_1 + I_2}$ for $I \in P$, and $B_h = \{I \in \mathbb{R}^2 : \max_i |I_i| < h\}$.

Definition 3.1 (Linear torus)**.** For $I \in P$, $K_I^0 : \mathbb{R}^2 \to E$ is the map of Convention 1.13.

Definition 3.2 (Analytic torus embedding)**.** A map $K : \mathbb{R}^2 \to E$ is an *analytic torus embedding* if it is $\mathbb{Z}^2$-periodic, real analytic (each point has a neighbourhood on which K is the restriction of a holomorphic map from a complex neighbourhood in $\mathbb{C}^2$ to the complexification $E_{\mathbb{C}}$), injective modulo $\mathbb{Z}^2$ ($K(\theta) = K(\theta') \Rightarrow \theta - \theta' \in \mathbb{Z}^2$), and has injective derivative $DK(\theta) : \mathbb{R}^2 \to E$ at every θ.

Definition 3.3 (Derivative along a frequency)**.** For $\nu \in \mathbb{R}^2$ and differentiable K, $D_\nu K(\theta) = DK(\theta)\nu$.

Definition 3.4 (Diophantine)**.** $\nu \in \mathbb{R}^2$ is (γ, τ)-Diophantine if $|k \cdot \nu| \geq \gamma |k|_1^{-\tau}$ for all $k \in \mathbb{Z}^2 \setminus \{0\}$, $|k|_1 = |k_1| + |k_2|$. (Same as the frozen track-K definition `KAM.IsDiophantine`.)

3.2. **Statement.**

Theorem 3.5 (W-NLW2, proposed)**.** *For every $m > 0$ there exist $r_0 > 0$, $C > 0$, $\tau > 1$, a function $\eta : (0, r_0) \to [0, \infty)$ with $\eta(r) \to 0$ as $r \downarrow 0$, a Borel set $\Gamma \subset \{I \in P : r(I) < r_0\}$, a continuous map $\nu : \Gamma \to \mathbb{R}^2$, and maps $K_I = (Q_I, P_I) : \mathbb{R}^2 \to E$ ($I \in \Gamma$) such that $(I, \theta) \mapsto K_I(\theta)$ is continuous on $\Gamma \times \mathbb{R}^2$ and:*

(1) (Density one in actions.) $\displaystyle\lim_{h \downarrow 0} \frac{\mathrm{Leb}_2(\Gamma \cap B_h)}{\mathrm{Leb}_2(P \cap B_h)} = 1.$

(2) (Two independent frequencies.) *For every $I \in \Gamma$ there is $\gamma_I > 0$ such that $\nu(I)$ is (γ_I, τ)-Diophantine, and $\nu(I) \to (\lambda_1, \lambda_2)/(2\pi)$ as $I \to 0$ in Γ.*

(3) (Small embedded tori.) *Each K_I is an analytic torus embedding, and for all $I \in \Gamma$*

$$\sup_\theta \|K_I(\theta) - K_I^0(\theta)\|_E \leq \eta(r(I))\, r(I), \qquad \sup_\theta \|K_I(\theta)\|_E \leq C\, r(I).$$

(4) (Exact invariance.) *For every $I \in \Gamma$, $\theta \in \mathbb{R}^2$ and $j \geq 1$,*

$$D_{\nu(I)} Q_{I,j}(\theta) = \lambda_j P_{I,j}(\theta), \qquad D_{\nu(I)} P_{I,j}(\theta) = -\lambda_j Q_{I,j}(\theta) - G_q(Q_I(\theta))_j.$$

(5) (PDE solutions.) *For every $I \in \Gamma$ and $\theta_0 \in \mathbb{R}^2$, the function $u(t,x) = U_m\big(Q_I(\theta_0 + t\nu(I))\big)(x)$ is real analytic in a neighbourhood of every point of $\mathbb{R} \times [0, \pi]$, is a classical solution of (1) on $\mathbb{R}$, satisfies $u_t(t, \cdot) = V_m\big(P_I(\theta_0 + t\nu(I))\big)$, and*

$$\sup_{t \in \mathbb{R},\, x \in [0,\pi]} \big(|u| + |u_x| + |u_t|\big) \leq C\, r(I).$$

(6) (Family injectivity.) *For all $I, I' \in \Gamma$ and $\theta, \theta' \in \mathbb{R}^2$, if $K_I(\theta) = K_{I'}(\theta')$ then $I = I'$ and $\theta - \theta' \in \mathbb{Z}^2$. Thus distinct parameters give disjoint tori, and each torus is embedded (with (3)).*

Remark 3.6 (Reading of the statement).

- *Quantifier order.* $r_0, C, \tau, \eta, \Gamma, \nu, K$ are chosen after m and before I and θ_0; γ_I may degenerate as $I \to 0$. No uniformity in m is claimed.
- *Invariance* in (4) is the coefficient system (2) along the linear flow on the torus (period-one angles, Convention 1.13); by Proposition 2.6(i) it yields (5) once the required time regularity is checked (W2). The cubic term is the physical force by (4).

- *Normalization independence.* The statement uses only the fixed real Hamiltonian, the standard actions $I_j = (q_j^2 + p_j^2)/2$ (through K_I^0) and period-one angles; it contains none of the printed P96-W constants. The frequency asymptotics in (2) are only the leading term; the twist A of Theorem 2.10 enters the proof, not the statement.
- *Family injectivity* (6) is proved in W8 from the source's family construction: the tori are images of the parameter tori under a single map that is injective on the union of the parameter tori (P96-W Main Theorem pp. 272–273, "embedding" of the family; Cantor Manifold Theorem and Remark 1 pp. 288–289).
- *Deviations from the source* (to be audited): pure cubic nonlinearity; $J = \{1, 2\}$; $a = p = 1$; period-one angles; Borel density-one set instead of a Cantor set (isolated points are allowed, cf. KP96 p. 24, Remark 2); linear stability, Whitney regularity, and the Lipschitz extension of P96-W Remark 2 p. 289 are omitted; the closeness to K_I^0 is a little-$o(r)$ bound rather than the $O(r^\sigma)$, $\sigma > 1$, of the radial map of the Cantor Manifold Theorem (P96-W p. 289), because the linearizing parametrization of Remark 1 there has a worse estimate (KP96 §7, Step 7).

Remark 3.7 (Declared weakenings). Relative to P96-W's Main Theorem specialized to $J = \{1, 2\}$, W-NLW2 declares the following weakenings (orchestrator ruling W-A2, recorded for owner review). *Deferred together* (D30 Q3): linear stability of the tori; Whitney regularity of the family; and the Lipschitz regularity of the family in Cartesian coordinates, i.e. the source's Lipschitz Cantor manifold (P96-W pp. 272–273, 288–289); W-NLW2 requires only continuity of $(I, \theta) \mapsto K_I(\theta)$ on $\Gamma \times \mathbb{R}^2$. *Retained*: each torus is an analytic embedding (3), and the family is injective (6), so the union of the tori is a disjoint union of embedded tori; what is not asserted is that the induced map on $\Gamma \times \mathbb{T}^2$ is a Lipschitz (bi-Lipschitz-parametrized) embedding. *Also declared*: Borel density-one parameter set instead of a Cantor set; $o(r)$ closeness instead of $O(r^\sigma)$; pure cubic nonlinearity, $a = p = 1$, period-one angles (Remark 3.6).

Source: P96-W Main Theorem pp. 272–273 (condition (4) holds for $J = \{1, 2\}$), Lemma 1 p. 278, Cantor Manifold Theorem and Remark 1 p. 289, assembly pp. 290–291; KP96 Theorem 2 p. 19 and §7 pp. 25–31; P96-K Theorems A, D.